

CASE STUDY

Individual Assignment

NSW DIGITAL AND DATA STRATEGY

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EXECUTIVE SUMMARY

The Data and Digital Strategy for the New South Wales Government builds on this bold, whole-of-government agenda to transform the use of data and digital technologies to deliver world-leading public services.

The strategy is set to make the Australian Public Service fit for this modern age, improving management, providing data, and delivering services digitally—all underpinned with increased cybersecurity. The focus will be on the data and digital architectural framework that accentuates the point of access, efficiency, and security for additional public trust and service excellence while ensuring it defines value from data as a national asset. These will include some of the major elements: The human-centric design is meant to enhance service accessibility and satisfaction of the user. The strong cybersecurity framework ensures the protection of the data integrity, while a constant commitment for improvement and innovation through the adoption of advanced technologies.

Besides, it collaborates with the public sector for the purpose of encouraging cooperation and influencing the integration of data-based decision-making processes. The strategy also seeks to develop a workforce of digital skills and effective change management practice for adapting to the changing digital landscape. This will frame the outline of the strategy, delivering world-class public services, seamless, secure, supporting trust and reliability for the people of New South Wales.

This is not only efficient in the operations of the government, but such an approach would even mean overall enhanced user experience that the public sector would be prepared to embrace and support. In other words, the balance measures the effectiveness of the strategy in its implementation, with technological advance and with the needs and expectations of citizens, so that digital transformation lines up with other wider goals of public service reform.

(Digital Transformation Agency, 2024)

Introduction: Background and Context of the Government Data Strategy.

Governments throughout the everywhere is innovating and adapting in order to provide services in an era of fast technological innovation and changing public expectations. Similar to other governments, the Australian government has realised how important it is to change the way it handles data and digital governance through the creation of a thorough plan. The Australian Public Service (APS) has been struggling with underfunding and outsourcing for almost ten years, as per the minister foreword which has forced it to re-evaluate its capacity to stay up with technology developments and adapt to the changing needs of businesses and residents.

With the goal of modernizing and improving the Australian Public Service as a foundation, the Australian government has started a reform path. In order to maximize data as a national asset, enhance digital service delivery, and strengthen cybersecurity, a strong Data and Digital Government Strategy must be developed and put into action. This plan arranges Australia to take advantage of possibilities, reduce risks, and promote public service trust in the future rather than just reacting to the problems of the present.

The Government Data Strategy is examined in this essay along with its history, context, goals, and effects on the Australian Public Service and society at large. Through an examination of this strategy's nuances, we hope to clarify its importance as a driving force behind revolutionary change and a model for creating a data-driven, digitally empowered government that is ready for the opportunities and challenges of the twenty-first century.

Identify Business Strategy and Digital Innovations:

STRATEGY FOR APS



(FIGURE 1)

This graphic is an educational tool divided into four main areas, each of which is intended to offer resources and advice about the Australian Government's Data and Digital Government Strategy.

What does the Strategy mean to me?

Individual public service roles are taken into consideration here, implying that everyone has a role to play—not only those in roles that are data-centric or ICT-focused. Public employees are advised to familiarize themselves with the goals and outcomes of the Strategy and to coordinate their level-data and digital initiatives with it.

How Will Success Be Calculated? Where can I find out more information?

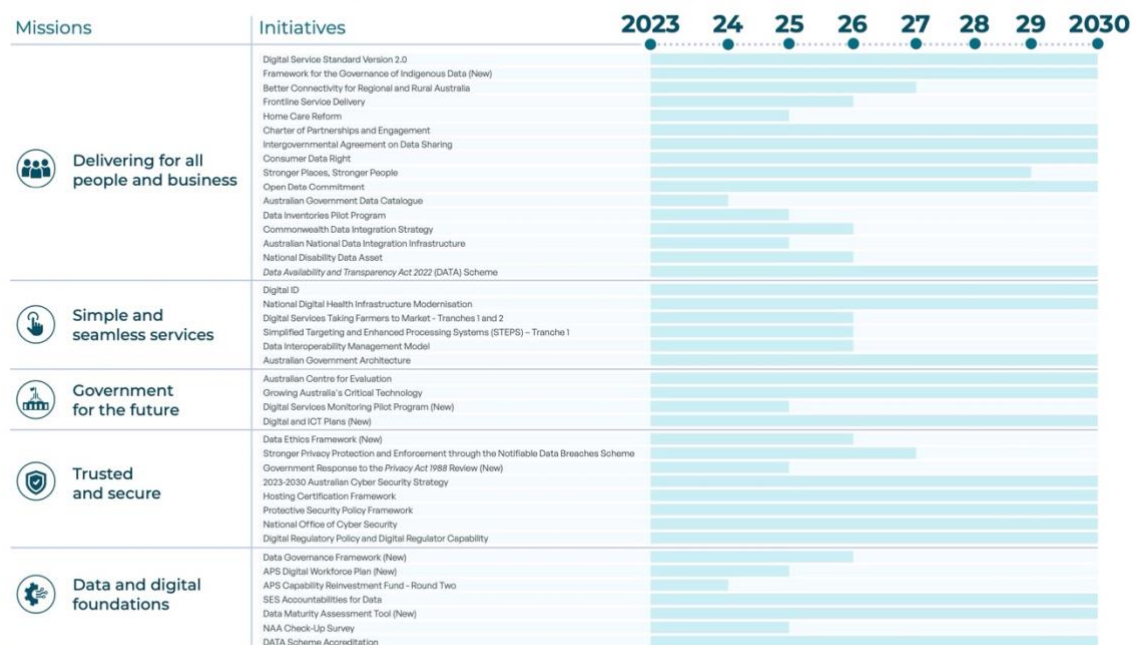
This section describes how the Strategy's performance will be measured by the Implementation Plan using current frameworks and metrics. By establishing connections with the APS Academy, APS professional groups, and data literacy initiatives, it also provides opportunities for additional education and learning. For individuals interested in further information, strategy leads' contact information is also provided.
(2023)

Thematic icons and graphics, such as people using computers or having conversations, are included in each area to represent cooperation and digital involvement. The document looks authoritative and approachable thanks to the professional color scheme, which combines white, blue, and teal. The Australian Government's larger data and digital strategy can be understood and engaged with using this infographic as a reference.

Implementation Plan

Roadmap

The roadmap captures a selection of new and existing data and digital initiatives across the APS which contributes to the Australian Government's data and digital agenda.



Footnote: The roadmap does not represent an exhaustive list of initiatives. It captures a sample of new and existing initiatives that align to the Strategy's 2030 vision.

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(FIGURE 2)

This picture is a of a roadmap that is a component of the Australian government's Data and Digital Government Strategy's implementation plan through 2030. The roadmap lists a number of current and upcoming projects that are in line with the digital and data agenda of the government, grouped according to the strategy's missions.

Actions and Missions:

Four main missions are outlined in the roadmap, each with corresponding initiatives:

1. Simple and seamless services: Projects like the Simplified Trade System and the modernization of digital health infrastructure fall under this category. They both seek to increase the effectiveness and efficiency of government services.

3. Government for the future: This mission aims to future-proof government services through the establishment of an Australian Centre for Evaluation and the improvement of the Australian Government Architecture, among other projects.

2030 VISION



(FIGURE 3)

The introduction, enablers, and missions make up the three primary sections of the graphic's framework.

Missions:

To realize the 2030 vision, the following three major missions are listed:

1. Fulfilling the needs of everyone and the business: This entails co-design, inclusivity, bolstering partnerships, and optimizing data value.

2. Straightforward and seamless services: Here, the idea of "tell us once" to cut down on duplication in information delivery to government services is combined with an emphasis on digital design, scalability, and a secure infrastructure.

(Edquist et al., 2022)

3. Future-oriented government: Embracing new technology and updating investment strategies are emphasized.

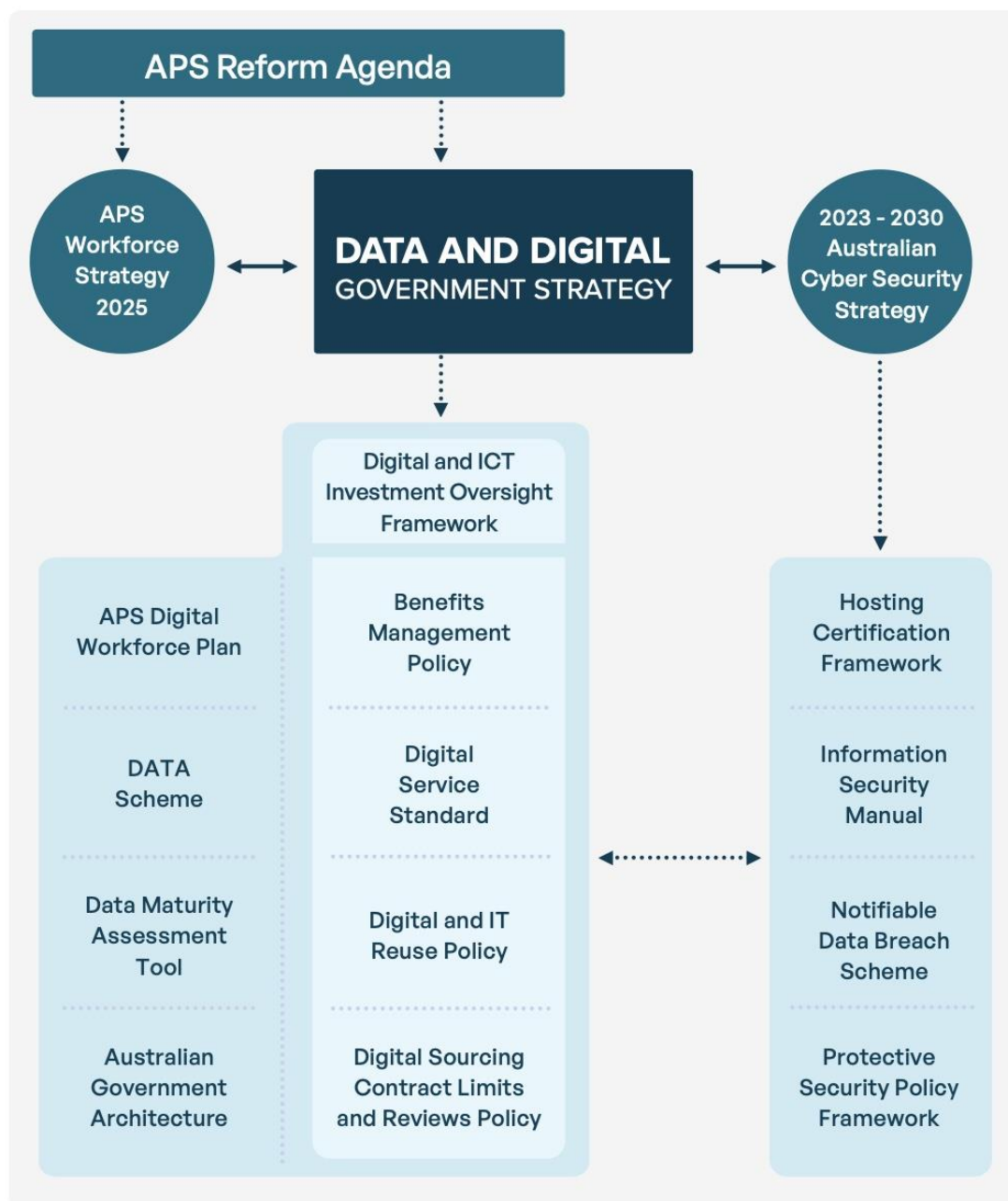
Enablers:

These are the fundamental components that the approach needs to succeed. Among them are:

Individuals' capacity

2. Administration

3. Investing in cyber security



(FIGURE 4)

The graphic displays a flowchart that, in relation to more general reform agendas and cybersecurity policies, graphically arranges the Australian government's Data and Digital Government Strategy.

Digital Innovations as Catalysts for Business Strategy

Within the Australian Government's Data and Digital Strategy, Maintaining this equilibrium is essential to the effective execution and approval of digital transitions. These elements are essential to guaranteeing that digital projects meet the expectations, needs, and capabilities of the general public as well as government workers, in addition to assuring their technical functionality.

1. Accessibility and Human-Centric Design

The plan focused on human-centric design is the core socio-technical elements. This methodology guarantees crafted with a profound comprehension of end-user requirements, encompassing accessibility for every person, irrespective of their physical capabilities or technological aptitude. To ensure that they serve a diverse spectrum of customers, including those with disabilities, all digital services, for example, must comply with the Digital Service Standard Version. This fosters inclusion, a fundamental principle in the provision of public services, in addition to improving customer pleasure. (Edquist et al., 2022)

2. Collaboration with Stakeholders and Co-Design

Stakeholder engagement is yet another essential socio-technical component. The government has implemented co-design procedures, which entail working together during the planning and execution stages of digital initiatives with the public, business leaders, and other relevant parties. Initiatives such as the Framework for the Governance of Indigenous Data, which was created in collaboration with Indigenous leaders to guarantee that data governance complies with the unique requirements and sensitivities of Indigenous communities, are clear examples of this approach. By this kind of involvement, digital solutions are made to be both socially acceptable and effectively solve the problems (Alex Antic, 2023)

3. Training and Capability of the Workforce

In addition to technology advancements, enhancing the data and digital competencies of public servants is crucial for the shift to a data-driven public sector. The goal of the APS Digital Workforce Plan and associated training programs is to give government workers the abilities they need to successfully adopt and operate new digital systems. In order to ensure that staff members are not just technically competent comprehend the wider ramifications of their work on privacy, security, and public trust, this involves training in data analytics, cybersecurity, and digital ethics. (Alex Antic, 2023)

4. Data Use Ethics

Data usage ethics are a major topic in the context of digital transformation. The Data Ethics Framework addresses concerns including accountability, transparency, and the public's influence on decisions made based on data. It also presents standards for moral decision-making in the public sector's use of data. In order to preserve the public's trust in government services, this framework helps guarantee that government agencies continue to be reliable custodians of public data. (Edquist et al., 2022)

5. Organizational Culture and Change Management

The socio-technical approach recognizes the value of change management techniques in assisting stakeholders and employees in adjusting to new working practices. In order to overcome opposition and promote an innovative and continuous improvement culture, this calls for the use of support networks, leadership engagement techniques, and communication tactics.

(Edquist et al., 2022)

To sum up, the Australian Government's Data and Digital Strategy highlights the importance of the socio-technical and soft components of business analytics in facilitating the successful integration of digital advances into public administration. These elements emphasize the organizational and human components, making sure that technical

developments support public service delivery rather than impede it. This promotes an inclusive, moral, and citizen-responsive government.
(APS 2023)

Metrics framework

	Outcomes	Metrics	Baseline	Why this measure?
	Embed co-design	% of citizens satisfied with government services	72% (June 2023) Source: Trust in Australian Public Services (TAPS) Survey	An indicator of well-designed, accessible and inclusive services.
	Embed inclusion and accessibility	% of users satisfied while engaging with government services	New metric to be piloted in 2023-24 Source: Digital Transformation Agency	An indicator of service satisfaction during use.
	Strengthen partnerships	% of services co-designed in line with the requirement of the Digital Service Standard	New metric to be established in 2023-24 to support new services from 1 July 2024, as per the Digital Service Standard Transition Plan Source: Digital Transformation Agency	New co-design metric to be considered as part of the Digital Service Standard metrics and evaluation framework in 2024
	Maximise value from data	# of data sharing agreements (including on Dataplace) # of organisations onboarded to Dataplace # of data assets discoverable and available for use	0 agreements on Dataplace (30 June 2023) Source: Office of the National Data Commissioner 47 organisations onboarded to Dataplace (30 June 2023) Source: Office of the National Data Commissioner 106,619 datasets on data.gov.au (December 2023) Source: data.gov.au	An indicator for the strength of partnerships between the Commonwealth and other entities. An indicator of organisations using and sharing of Australian Government data via the Dataplace platform. An indicator for the publication and availability of Government data as open data. Release of public data as open data has been shown to have productivity and economic benefits.
	Be digital by design	% of services that are started digitally and successfully completed (digital service completion rate)	New metrics to be trialled in 2024 Source: Digital Transformation Agency	Digital completion measures the number of users that start and complete the transaction.
	Deploy scalable and secure architecture	# of users accessing interoperable architecture for services – e.g. myGov, myGovID	Linked Active Accounts Access in 24 months as at 30 June 2023: 21,577,305 Linked to more than 2 Commonwealth services – Access within 24 months: 12,664,498 Linked to at least 1 state service – Accessed within 24 months: 255,221 Source: Services Australia Digital Program Management Office (Department of Finance)	The number of myGov accounts linked to services indicates the number of Australians actively accessing government digital services.
	Tell us once	% of time a service is available to users % of people reporting having to provide the same information to more than one service	New metrics to be trialled in 2024 Source: Digital Transformation Agency 64% of people agreed with the statement, "Information I provided to one service didn't need to be repeated to others" (2022-23 financial year) Source: Trust in Australian Public Services (TAPS) Survey	An indicator of the availability of a service. An indicator of simple and seamless services, and the extent to which the public agrees that the "Tell us once" approach is working
	Adopt emerging technologies	% of digital investment proposals in-scope of the Investment Oversight Framework (IOF) which meet the requirements of the DTA's Benefits Management Policy as assessed by the DTA in a financial year.	New metric to be trialled in 2024 to establish a baseline Source: Digital Transformation Agency	Strengthening benefits articulation, tracking, and reporting is key to driving improved outcomes on digital investments and provides a clear line of sight from digital investments.
	Modernise investment approaches			
	Build and maintain trust	% people who trust government services	61% of people (June 2023) Source: Trust in Australian Public Services (TAPS) Survey	This measures the extent to which users trust the service they are using. Interactions with services can be a proxy for broader issues of trust in government.
	Modernise legislation	Entity security maturity rating	83% of entities manage compliance with all Protective Security Policy Framework policies (2022) Source: Protective Security Policy Framework Reporting	An entity security maturity assessment helps to identify areas for improvement and benchmarks against peers.
	Connect data, digital and cyber security			
	Manage data as a valuable national asset	% of current APS workforce in data and digital jobs	10.6% of APS Jobs (December 2022) Source: Australian Public Service Employment Release (APSER) of select job families as they relate to data or digital	An indicator of the proportion of APS employed in data and digital job roles. This measure provides visibility of the APS data and digital workforce and whether it changes over time.
	Build a data and digital-capable APS	% of women in data and digital roles in the APS workforce	41% of data and digital roles in the APS are filled by women (December 2022) Source: Australian Public Service Employment Release (APSER) of select job families as they relate to data or digital	An indicator of the proportion of women employed in APS data and digital job roles.
	Grow APS maturity	% APS employee satisfaction	74% of APS employees (20 June 2023) Source: APS Central – Australian Public Service Commission	An indicator of the proportion of APS employee satisfaction. Employee satisfaction is correlated with customer satisfaction, engagement and productivity.
		Overall entity Data Maturity Rating	New metric to be trialled in 2024 Source: Department of Finance	This measure will benchmark the data maturity of entities over time. It is based on the Data Maturity Assessment Tool currently under development, and will help inform future action where it is needed most.

 Delivering for all people and business
  Simple and seamless services
  Government for the future
  Trusted and secure
  Data and digital foundations

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(FIGURE 5)

A thorough "Metrics Framework" has three primary columns: outcomes, metrics, and baseline. There's also a fourth column called "Why this measure?" that provides a justification for the metrics selected.

(2023)

Socio-Technical and Soft (Non-Digital) Aspects

Sociotechnical aspects of business analytics:-

Human Factors: Human interpretation and insight-based decision-making are crucial elements of business analytics, in addition to data and algorithms. Understanding people's perspectives and usage of analytics is crucial.

Organizational Culture: An organization's adoption and application of analytics are influenced by its culture. Effectiveness is impacted by elements like the importance given to analytical abilities and receptivity to data-driven judgments.

Team Dynamics: Cross-functional teams are frequently necessary for analytics. Leveraging collective intelligence requires an understanding of communication and team dynamics.

Implications for Ethics and Society: The use of data gives rise to issues related to prejudice and privacy. Analytics procedures must be in line with legal requirements and moral principles within organizations.

Transformation Management: Organizational transformation may be necessary to implement analytics. Addressing opposition and offering support are essential components of effective change management.

SOFT SKILL ASPECTS

- 1. Understanding User Requirements and Expectations:** Crucial for designing analytics solutions that cater to user needs and yield valuable insights.
- 2. Effective Communication and Collaboration:** Crucial among stakeholders, data scientists, and analytics teams to ensure alignment with business objectives and deliver actionable insights.
- 3. Managing Organizational Change:** It is critical for garnering stakeholder support and adoption of analytics solutions, significantly impacting the success of initiatives.

4. **Interpreting and Conveying Results:** Conveying accurate results is vital to present analytics model results in a comprehensible and actionable manner for business users, influencing the value derived.
 5. **Addressing Ethical and Legal Considerations:** This is important for maintaining trust and acceptance of analytics solutions by addressing data privacy, security, and compliance issues.
 6. **Holistic Perspective:** Embracing a socio-technical systems approach acknowledges the intricate interplay between technology, individuals, and processes. By grasping these interconnected elements, organizations can craft government strategies that resonate with stakeholders' needs and behaviors.
 7. **Interdisciplinary Collaboration:** Fostering collaboration across technical experts, policymakers, and end-users ensures that data and digital solutions are not only technically proficient but also mindful of social implications and user needs.
 8. **Change Management:** Considering socio-technical aspects aids in navigating organizational changes effectively. Implementing data and digital government strategies often entails shifts in processes, roles, and workflows, all of which can be better managed by understanding the social dynamics within the organization.
- (2023)

Here's how socio-technical and soft dimensions contribute to shaping data and digital government strategies:

1. **User-Centric Design:** Developing user-friendly data and digital government solutions requires giving priority to soft factors like user experience (UX) and human-centered design concepts. Organizations can increase adoption and usefulness by focusing on the requirements and preferences of their users.
2. **Stakeholder Engagement:** Putting a strong emphasis on including stakeholders in the strategy-development process makes it easier to obtain information, build consensus, and make sure that solutions meet the needs of a variety of stakeholders.

3. **Ethical Aspects:** Soft aspects include moral issues including security, privacy, and openness of data. Strategies that incorporate ethical standards are more likely to build stakeholder confidence.

Conclusion and Recommendations:

With an emphasis on efficiency, equality, and accessibility, the Australian Government's Data and Digital Strategy successfully integrates digital advances. The plan skilfully synchronizes digital tools with organizational goals, improving service quality and customer experience. In order to improve the method even more, the following suggestions are put forth:

1. **Strengthen Cybersecurity Initiatives:** To protect sensitive data and maintain public confidence, strengthen cybersecurity measures. It is imperative that security measures be updated on a regular basis and that all government employees receive thorough cybersecurity training.
2. **Expand Digital Literacy Programs:** To guarantee that all residents may equally benefit from digital services, expand digital literacy programs, especially for underprivileged and isolated populations. This will improve service accessibility and aid in bridging the digital divide.
3. **Improve Data Privacy Protocols:** To meet the growing privacy concerns, strengthen data privacy rules. The public's trust in government services will be strengthened by transparent data management techniques.
4. **Sustain Active Stakeholder Engagement:** Keep up a dialogue with stakeholders by working together on designs and providing opportunities for feedback on a frequent basis. By doing this, it will be possible to guarantee that digital services can adjust to changing consumer demands and expectations.
5. **Emphasis on Ecological Technology Approaches:** Adopt green practices and technologies to incorporate sustainability into the digital strategy. This involves reducing the environmental effect of digital operations by selecting sustainable technology solutions and optimizing data centers for energy efficiency.
6. **Boost Vulnerable Populations' Accessibility:** Continue to customize digital services to the unique requirements of vulnerable groups, such as the elderly, those with impairments, and non-

native speakers. This can entail different access points, support services, or customized interfaces.

7. **Create Future-Oriented Training Programs:** To keep up with the quick changes in technology, public servants should invest in future-oriented training programs. Training in blockchain, AI, and other cutting-edge technologies that may have an influence on service delivery in the future is part of this.
8. **Reinforce New Technologies-Related Legal Frameworks:** Revise legislative structures to take new technologies' difficulties into account. Clarifying data ownership, the ethical use of big data, and the legal ramifications of AI judgments in public services are some examples of this.

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